Nikhilesh Gawhale

Generative AI / Machine Learning Engineer | RAG, LLM Fine-Tuning and Production ML Pipelines
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PROFESSIONAL SUMMARY

Generative AI / Machine Learning Engineer with an MSc in Artificial Intelligence (Distinction) from Queen Mary University of London and three years of hands-on experience designing, building and productionizing GenAI systems end to end. Eighteen months building Retrieval-Augmented Generation (RAG) pipelines, LLM-based architectures and agentic workflows for a regulated financial services platform, including fine-tuning pre-trained language models for domain-specific use cases. Strong Python, solid understanding of ML/GenAI evaluation techniques, and experience building data pipelines and deploying models to production on AWS. Comfortable owning a problem end to end with high independence, in a fast-moving environment with real business outcomes to hit.

TECHNICAL SKILLS

Generative AI and LLM Architectures: RAG and LLM-based architectures in production - ingestion, chunking strategy, embeddings, hybrid retrieval, re-ranking; transformers and modern generative AI techniques; prompt engineering and structured outputs; Anthropic Claude, OpenAI, Azure OpenAI, AWS Bedrock
LLM Fine-Tuning: Fine-tuning pre-trained language models for domain-specific and task-specific use cases using LoRA on a 10,000-document enterprise corpus - dataset preparation, training and held-out evaluation
ML/GenAI Evaluation: Solid evaluation technique - labelled gold sets, per-class precision, recall and F1, LLM-judge scoring, calibration; continuous integration regression testing; benchmarking model performance and quality rather than relying on a single aggregate figure
Agentic Systems: Multi-step agent workflows with tool calling, memory and orchestration; LangGraph, LangChain, CrewAI; Model Context Protocol (MCP) tool integrations; proof-of-concept GenAI systems built and deployed independently
Engineering and Production: Strong Python (asyncio, FastAPI, Pydantic v2, pytest); building and maintaining data pipelines for training and inference workflows; Git, Docker, Linux-based systems; software engineering practices applied to ML and GenAI pipelines
Cloud and MLOps: AWS (Bedrock, Lambda, S3) and Microsoft Azure (OpenAI, AI Search, AKS); continuous integration and continuous delivery (CI/CD); monitoring deployed models for performance, reliability and quality drift; PostgreSQL, MongoDB, Redis, SQL

PROFESSIONAL EXPERIENCE

AI Engineer, RP-Tech International Solutions (Investment Banking Technology), Remote | January 2025 - August 2026

- Designed, built and productionized Generative AI and agentic systems for a regulated financial services platform, owning the pipeline end to end from training and fine-tuning through deployment and monitoring.
- Built RAG pipelines and LLM-based architectures over large volumes of unstructured documents: ingestion, chunking strategy, embeddings, hybrid retrieval and re-ranking, with grounding checks verifying outputs against source text.
- Fine-tuned pre-trained language models using LoRA on a 10,000-document financial corpus for domain-specific performance, covering dataset preparation, training and held-out evaluation.
- Built and maintained data pipelines for training and inference workflows, and deployed proof-of-concept GenAI systems that progressed into production use; evaluated and benchmarked model performance with a 500-document labelled corpus and per-class metrics, raising accuracy from 70 percent to 80 percent.
- Monitored deployed models in production and continuously improved reliability, deploying via Docker and Kubernetes on AWS and Azure with full observability for latency, throughput and cost.
- **Generative AI Engineer, Midas Advisory, London, United Kingdom | August 2024 - December 2024**
- Fine-tuned language models using LoRA on a corpus of over 10,000 financial documents; built RAG pipelines and LLM features in Python and FastAPI, deployed via Docker and CI/CD.

PROJECTS

Underwriter Copilot, Python, Azure OpenAI, Azure AI Search, MongoDB, event-driven services | github.com/Nikhileshgawhale/underwriter-copilot

- Built solo, end to end: an agentic system for insurance policy review with clause comparison, deviation flagging and precedent retrieval, evaluated on per-class F1 and retrieval precision rather than a single accuracy figure.

EDUCATION

MSc Artificial Intelligence - Distinction, Queen Mary University of London, United Kingdom | September 2023 - September 2024

BCA Bachelor of Computer Applications - GPA 7.8/10, Shivaji Science College, Amravati, India | August 2019 - August 2022